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Subject: OSSP(PRACTICAL)

1. PIPES

```
vaibhavi@VYSRI: ~/pipes
GNU nano 8.7.1
#include <stdio.h>
#include <sys/types.h>
#include <sys/stat.h>

int main()
{
    if (mkfifo("myfifo", 0666) == -1)
    {
        perror("mkfifo");
        return 1;
    }

    printf("FIFO created\n");
    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI: ~/pipes
vaibhavi@VYSRI:~$ mkdir pipes
vaibhavi@VYSRI:~$ cd pipes
vaibhavi@VYSRI:~/pipes$ nano fifo_create.c
vaibhavi@VYSRI:~/pipes$ gcc fifo_create.c -o fifo_create
vaibhavi@VYSRI:~/pipes$ ./fifo_create
FIFO created
vaibhavi@VYSRI:~/pipes$ ls
fifo_create  fifo_create.c  myfifo
vaibhavi@VYSRI:~/pipes$ |
```

2.FIFO_READ

```
vaibhavi@VYSRI: ~/pipes
GNU nano 8.7.1
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int fd;
    char buffer[100];

    fd = open("myfifo", O_RDONLY);
    read(fd, buffer, sizeof(buffer));
    printf("Received: %s\n", buffer);
    close(fd);

    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI: ~/pipes
vaibhavi@VYSRI:~/pipes$ cd ~/pipes
vaibhavi@VYSRI:~/pipes$ nano fifo_read.c
vaibhavi@VYSRI:~/pipes$ gcc fifo_read.c -o fifo_read
vaibhavi@VYSRI:~/pipes$ ./fifo_read
```

3.FIFO_WRITE.C

```
vaibhavi@VYSRI: ~/pipes
GNU nano 8.7.1                                fifo_write.c
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <string.h>

int main()
{
    int fd;
    char message[] = "Hello from Writer";

    fd = open("myfifo", O_WRONLY);
    write(fd, message, strlen(message) + 1);
    close(fd);

    printf("Message sent\n");
    return 0;
}
```

OUTPUT:

```
vaibhavi@VYSRI: ~/pipes
vaibhavi@VYSRI:~/pipes$ cd ~/pipes
vaibhavi@VYSRI:~/pipes$ nano fifo_write.c
vaibhavi@VYSRI:~/pipes$ gcc fifo_write.c -o fifo_write
vaibhavi@VYSRI:~/pipes$ ./fifo_write
Message sent
vaibhavi@VYSRI:~/pipes$ |
```

```
vaibhavi@VYSRI: ~/pipes × Ubuntu × +  
vaibhavi@VYSRI:~$ cd ~/pipes  
vaibhavi@VYSRI:~/pipes$ ./fifo_read  
Received: Hello from Writer  
vaibhavi@VYSRI:~/pipes$ █
```